IIITB COMET FOUNDATION

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Abstract

Simulation of latch behavior using Raspberry Pi Pico to demonstrate NAND and NOR latch transitions for the input combinations (0,1) → (1,1).

1. Components

Component	Qty
Pico2w	1
Push Buttons	2
LEDs	2
220Ω Resistors	4
Breadboard	1
Jumper Wires	10
Laptop with Thonny IDE	1

Table: Components used

2. Setup

- GP15: Input P1 (Push Button)
- GP14: Input P2 (Push Button)
- GP16: NAND Q Output (LED)
- GP17: NOR Q Output (LED)
- GND and VBUS properly connected

3. Observation

- **NAND Latch:** (0,1) → (1,0) → holds at (1,0)
- **NOR Latch:** (0,1) → (1,0) → transitions to (0,0)

4. Truth Tables

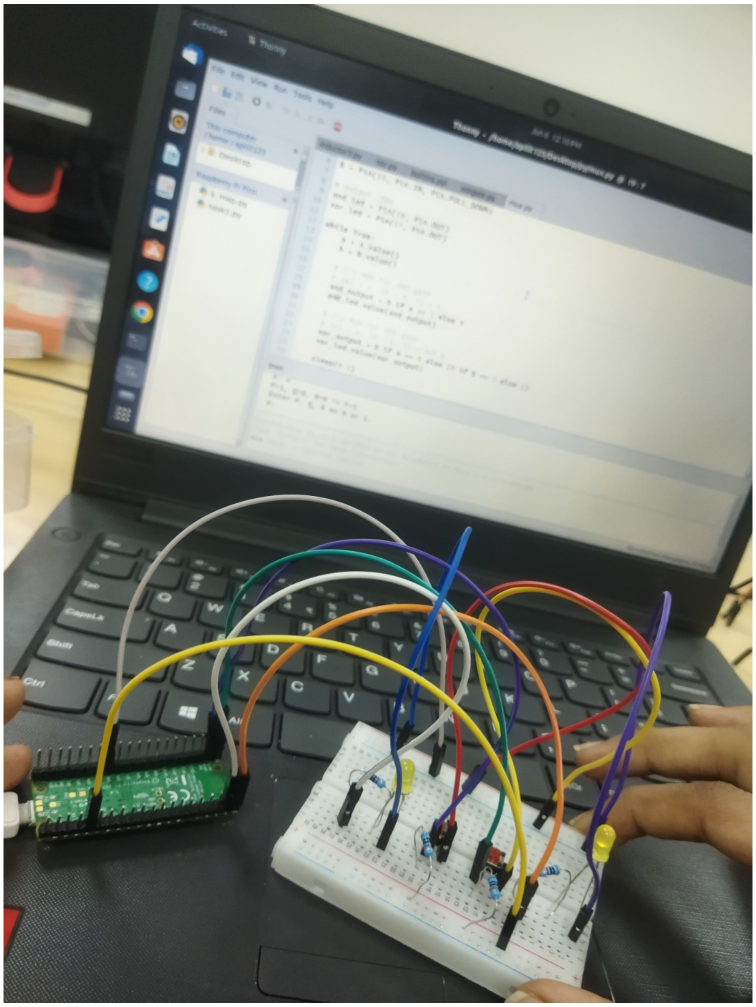
NAND Latch

$P1$	$P2$	Output (Q1, Q2)
0	1	$(1,0)$
1	1	$(1,0)$ (holds)

NOR Latch

$P1$	$P2$	Output (Q1, Q2)
0	1	$(1,0)$
1	1	$(0,0)$

5. Circuit Image



6. Procedure Steps

1. Connect Raspberry Pi Pico to a breadboard.
2. Interface push buttons to GP14 and GP15 using 10kΩ pull-down resistors.
3. Connect LEDs with 220Ω resistors to GP16 (NAND Q) and GP17 (NOR Q).
4. Ensure all GNDs are properly tied together.
5. Open Thonny IDE and write or upload MicroPython latch code (from GitHub).
6. Run the script on the Pico and observe outputs for different button combinations.
7. Compare actual LED outputs with expected latch behavior as per truth tables.

7. GitHub Repository

- Source files and documentation available at:
  `https://github.com/Rithrishareddy/RITHRISHA_FWC/tree/main/Hardware`

8. Conclusion

This project successfully demonstrates the basic latch behavior for NAND and NOR gates using the Raspberry Pi Pico. The state transitions are observed clearly through LED indicators based on push button inputs, validating digital memory concepts.